

MA-105 Tutorial-7

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August 2025

1. If $f(u, v, w)$ is differentiable and $u = x - y$, $v = y - z$, $w = z - x$, show that

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} = 0$$

2. Given $w = 2ye^x - \ln z$, $x = \ln(t^2 + 1)$, $y = \tan^{-1} t$, $z = e^t$, find $\frac{dw}{dt}$ at $t = 1$.
3. Find the directions in which the following functions increase or decrease most rapidly at the given point P_0 , then find the derivative of the function in those directions.

(a) $f(x, y) = x^2 + xy + y^2$, $P_0 = (-1, 1)$

(b) $f(x, y) = x^2y + e^{xy} \sin y$, $P_0 = (1, 0)$

(c) $f(x, y, z) = \frac{x}{y} - yz$, $P_0 = (4, 1, 1)$

(d) $g(x, y, z) = xe^y + z^2$, $P_0 = (1, \ln 2, \frac{1}{2})$

(e) $f(x, y, z) = \ln xy + \ln yz + \ln zx$, $P_0 = (1, 1, 1)$

(f) $h(x, y, z) = \ln(x^2 + y^2 - 1) + y + 6z$, $P_0 = (1, 1, 0)$

4. Find the points on the curve $x^2 + xy + y^2 = 1$ in the xy plane which are nearest to and farthest from the origin.
5. Find the points of maxima, minima and saddle point of the following curves using Hessian matrix:
 - (a) $f(x, y, z) = x^2 - 4y^2 + 2z^2 - 2xy + 6yz + 6xz$
 - (b) $f(x, y, z) = 2xyz - 4xz - 2yz + x^2 + y^2 + z^2 - 2x - 4y + 4z$
6. Sketch the following curves:
 - (a) $x^2 + y^2 + z^2 = 1$ (Sphere)
 - (b) $x^2 + y^2 - z^2 = \alpha^2$ ($\alpha \neq 0, \alpha \in \mathbb{R}$)
 - (c) $x^2 - y^2 - z^2 = \alpha^2$ ($\alpha \neq 0, \alpha \in \mathbb{R}$)
 - (d) $x^2 + y^2 = z^2$ (Cone)
 - (e) $x^2 + y^2 = z$ (Paraboloid)
 - (f) $x^2 - y^2 = z$ (Hyperbolic Paraboloid)